Healthcare Patient Management System — Dataset Documentation

UCI Machine Learning Repository: Diabetes 130-US Hospitals (1999–2008)

1. Dataset Overview & Source

Dataset Name: Diabetes 130-US Hospitals for Years 1999-2008
Origin: UCI Machine Learning Repository (Center for Clinical and Translational Research, VCU)
URL: https://archive.ics.uci.edu/dataset/296/diabetes+130-us+hospitals+for+years+1999-2008
Scope: Over 100,000 hospital encounters collected across 130 United States clinical facilities over a 10-year observational timeline.

2. Clinical Relevance & Key Attributes

The dataset tracks diabetic patient encounters and pharmacological interventions to evaluate outcomes and unplanned hospital readmissions. It provides critical operational and clinical metrics:

Attribute	Data Type	Clinical Description
encounter_id	Integer (PK)	Unique administrative identifier for the inpatient hospital encounter
patient_nbr	Integer (FK)	Unique identifier for the patient (supporting longitudinal tracking across multiple admissions)
race / gender / age	Categorical	Demographic stratification (Caucasian, African American, Hispanic, etc.; 10-year age brackets)
admission_type_id	Integer (FK)	Category of admission (Emergency, Urgent, Elective, Newborn, Trauma)
discharge_disposition_id	Integer (FK)	Discharge destination (Discharged to home, Transfer to facility, Expired)
time_in_hospital	Integer (1-14)	Inpatient length of stay measured in whole days
num_lab_procedures	Integer	Total diagnostic laboratory procedures performed during the encounter
num_medications	Integer	Distinct pharmacological medications administered to the patient
diag_1, diag_2, diag_3	Categorical	Primary, secondary, and tertiary ICD-9 diagnostic codes
readmitted	Categorical	Primary hospital outcome (<30 days readmission, >30 days, or NO readmission)

3. Preprocessing & Data Cleaning Applied

- **Missing Value Normalization:** Converted raw '?' placeholders into structured database NULL values.
- **High-Missing Column Pruning:** Columns with >80% missing records (e.g. weight, payer_code in raw feeds) were isolated to avoid schema pollution.
- **ICD-9 Normalization:** Extracted primary, secondary, and tertiary diagnoses into a normalized 1-to-many relationship (`encounter_diagnoses`).
- **Medication Normalization:** Transformed 24 wide medication columns (insulin, metformin, etc.) into an active treatment junction table (`encounter_medications`).